

Université Paris-Dauphine – PSL University**Bureau P121**

LEDa – CGEMP

Place du Maréchal de Lattre de Tassigny

75775 Paris Cedex 16, France

+33 6 45 50 97 27

arthur.thomas@dauphine.psl.eu

<https://arthurthomaseconometrics.github.io>**EMPLOYMENT**

- 2022- **Assistant Professor (Maître de Conférences)**, *Université Paris-Dauphine, France*
- 2021- **Associated Researcher** in the Climate Economics Chair, *Université Paris-Dauphine, France.*
- 2017- **Associated Researcher** in the Chair of the Economics of Natural Gas, *Paris Dauphine-PSL University, IFP School, Mines Paris Tech-PSL University, and Toulouse School of Economics. The German Institute for Economic Research (DIW).*

VISITING RESEARCHER

- OCT. 2024 **Visiting Professor**, *Department of Economics at York University, Toronto, Canada. Invited by Prof. Jasiak*
- FEB-APR. 2024 **Visiting Professor**, *Department of Quantitative Economics at Maastricht University, School of Business and Economics, Netherland. Founded by University Paris Dauphine Grant for « Jeunes Chercheurs et Chercheuses ». Invited by Prof. Hecq*

GRANTS AND AWARDS

- 2024-2025 **PSL Global Seed Fund. Partners:** *University Paris Dauphine PSL, Maastricht University (Alain Hecq), Tor Vergata University of Rome (Gianluca Cubbada) and York University, Toronto (Joann Jasiak).*
- 2023-2024 **Paris-Dauphine Grant for Jeunes Chercheurs et Chercheuses**

EDUCATION

- 2017 - 2020 **PhD in economics**, Nantes University, France
Title: “The Econometrics of Energy Demand: Identification and Forecast”, under the supervision of Benoît Sévi (Nantes-University) and Olivier Massol (IFP School and City, University of London)
Committee: K. Abadir (Imperial College London, referee), D. Bunn (London Business School), D. Korobilis (University of Glasgow, referee), O. Massol (IFP School and City, University of London, supervisor), V. Mignon (Université Paris-Nanterre, chairman), B. Sévi (Nantes University, supervisor)
- 2014 - 2017 **M.Sc. in Statistics**, École Nationale de la Statistique et de l'Analyse de l'Information (ENSAI), France
- 2012 – 2014 **Preparatory classes (MP)**, Lycée Henry Poincaré (Nancy), France

PUBLICATIONS

Bruguet, M., Thomas, A., & Le Saout, R. (2025). Weather Effects in Energy Seasonal Adjustment: An Application to France Energy Consumption. *The Energy Journal*, 0(0).

Bardon P., Massol O., Thomas A. (2025). Greening Aviation with Sustainable Aviation Fuels: Insights from decarbonization scenarios. *Journal of Environmental Management*, vol. 374.

Thomas, A., Massol, O., & Sévi, B. (2022). How are Day-ahead Prices Informative for Predicting the Next Day's Consumption of Natural Gas? Evidence from France. *The Energy Journal*, 43(5), 1-26.

WORKING PAPERS

Gilles De Truchis Fries Sébastien, Arthur Thomas (2025). Forecasting extreme trajectories using semi-norm representations. [[Paper](#)][[Working paper](#)]

Gilles De Truchis Fries Sébastien, Arthur Thomas (2025). Prediction of bubbles in presence of α -stable aggregates moving averages.

Yannick Le Pen, Zakaria Moussa, Arthur Thomas (2025). Regime Switching for Dynamic EquiCorrelation. [[Paper](#)]

Olivier Massol, Eduard Civel, Arthur Thomas (2025). Unfair Trade in the Circular Economy? Price Dynamics in Chinese and European Waste to Biofuel Industries (2025). [[Working paper](#)]

Gilles De Truchis, Elena Dumitrescu, Sébastien Fries, Arthur Thomas (2024). Bet on a bubble asset? An optimal portfolio allocation strategy. [[Paper](#)]

Zakaria Moussa, Arthur Thomas (2024). Identifying Oil Supply News Shocks and Their Effects on the Global Oil Market, USAEE Working Paper No. 21-490, Jun. 26, 2023. [[Working paper](#)]

Zakaria Moussa, Benoît Sévi, Arthur Thomas (2021). Real-time demand in U.S. natural gas price forecasting: the role of temperature data, USAEE Working Paper No. 21-507, 21 Sep 2021. [[Working paper](#)]

ONGOING WORK

Olivier Massol, Eduardo S. Marques, Arthur Thomas (2025). Disentangling Drivers of EU Allowance Prices: A Mixed Causal and Non-Causal Time Series Approach.

Gilles De Truchis, Arthur Thomas, Ludvine Vaudree (2025), Deconvolution and Filtering of Non-Causal Alpha-Stable Processes.

Elena Dumitrescu, Julien Peignon, Arthur Thomas (2025). Tail-aware density forecasting of locally explosive time series: a neural network approach.

Gilles De Truchis, Fries Sébastien, Arthur Thomas (2025). Multivariate seminorm representation for α -stable moving average processes and path prediction.

REFeree REPORTS

25 reports since 2018 in the following journals: *Annals of Economics and Statistics*; *Empirical Economics*; *The Energy Journal*; *Energy Economics*; *European Economic Review*; *Journal of Times Series Analysis*; *Journal of Banking and Finance*; *Nature Energy*; *Resource and Energy Economics*; *Revue Economique*; *Revue Française d'Economie*; *Revue Française d'Economie Politique*.

ACADEMICS RESPONSIBILITIES

- 2025 **Organizer of the 1st International Noncausal Econometrics workshop.** This workshop was held on June 13th and 14th 2025 in Paris. Co-organized with the University of Orléans and University Paris II Panthéon-Assas. [[Program](#)]
- 2025 **Organizer of an invited session at 19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics),** CFE-CMStatistics 2025, Birkbeck, University of London.
- 2025 **Organizer of the 43rd IAEE International Conference,** Paris, France
- 2025 **External member in a recruiting committee for Assistant professor** (sections 5 and 26) AMSE, Marseille
- 2024 **Member of recruiting committee for Assistant professor** LEDa, Dauphine -PSL
- 2023-2025 **Organizer of the LEDa seminar ([link](#)),** Dauphine -PSL
- 2023 **Member of the working group on causality,** CEREMADE, LAMSADE, LEDa, Dauphine-PSL
- 2023,2025 **Organizer of the Journée portes ouvertes (JPO) du LEDa,** Dauphine -PSL
- 2023 **Organizers of the Annual Doctoral Conference of the Association for the Development of Research in Economics and Statistics (ADRES),** Dauphine-PSL
- 2023-2024 **Organizer of a LEDa cross-disciplinary seminar on theoretical quantitative techniques ([Econometrics seminar link](#)),** Dauphine-PSL
- 2020 **Organizers of the 37th International Conference of the French Finance Association (AFFI),** Nantes, France.
- 2018 - **Member of the "[Les Jeunes Economètres](#)" working group**

SELECTED CONFERENCES AND WORKSHOPS

- 2025 **13th Annual Conference of the International Association for Applied Econometrics (IAAE) 2025,** University of Torino, Torino.
- 46th IAEE International Conference 2025,** Paris France.
- 17th Annual SoFiE Conference,** ESSEC Campus of Cergy, France.
- 7th edition of QFFE 2025,** AMSE, Marseille, France.
- 15th Workshop in Time Series Econometrics,** Zaragoza.
- 18th Financial risk international forum,** Institut Louis-Bachelier .Paris.
- 2024 **Econometrics seminar series,** York university, Toronto.

	18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics), King's College London, Invited session. 39th Annual Meeting of the Canadian Econometrics Study Group (CESG), York university, Toronto. Séminaire FDD-FiME, Institut Henri Poincaré. 4th Italian Workshop of Econometrics and Empirical Economics, Free University of Bolzano, Italy. Maastricht Econometric Seminars <i>Department of Quantitative Economics at Maastricht University, School of Business and Economics, Netherland.</i> Internal seminar LEDa, University Paris Dauphine-PSL, France. 6th edition of QFFE 2024, AMSE, Marseille, France 44th International Symposium on Forecasting, Dijon France
2023	Workshop on noncausal models, Maastricht University, Netherland CREST Finance seminar, ENSAE, France 17th International Conference on Computational and Financial Econometrics, HTW Berlin, University of Applied Science, Germany, invited session. 22ème Journée d'Économétrie – Développements Récents de l'Econométrie Appliquée à la Finance, Nanterre, France
2022	FiME Summer School on “Big Data and Finance”, University Paris Dauphine, CREST, Institut Louis-Bachelier, France Internal seminar LEDa, University Paris Dauphine-PSL, France Big Data and Econometrics seminar, AMSE, France 16th International Conference on Computational and Financial Econometrics, King's College London, UK 21ème Journée d'Économétrie – Développements Récents de l'Econométrie Appliquée à la Finance, Nanterre, France Séminaire de recherche du LEO, Orléans, France 4th edition of QFFE 2022, AMSE, Marseille, France Research seminar of the ACSS-PSL Institute
2021.	7th RCEA Times series workshop, University of Milano-Bicocca.

PHD STUDENTS

2025 –	Aymen Bahrouni, PhD student at Dauphine, co-supervised with P. Geoffron. “De la finance verte à la formation du prix de l'électricité”, PhD funded by the LEDa, University Paris Dauphine-PSL.
2025 –	Julien Peignon, PhD student at Dauphine, co-supervised with F. Rossi (CEREMADE). “Improving the prediction of extreme events with machine learning, signal processing and non-causal econometrics” funded by the ENS Paris – Saclay. PhD in both mathematics and economics.

2023 – **Marie Bruguët**, *PhD student at Dauphine, co-supervised with A. Creti, “Measuring and evaluating sufficiency and efficiency in French residential energy consumption”, COFRA funded by the French Ministry of Ecological Transition.*
[\[website\]](#)

SUPERVISED MASTER STUDENTS (SINCE 2022)

2023. Lux, Boudon (M2-QEA), research assistant funded by Dauphine -PSL Graduate Program, jointly with F. Tripier (Univ. Dauphine) and Z. Moussa (Univ. Nantes), *Noncausality, monetary shocks, prices, wages, and profits.*

Janna Bengouirah (M2-QEA), *Noncausal portfolio optimization.*

Eddy Darragi (M2-QEA), *estimating residential consumption elasticities prices under subsidies in time of crisis*, jointly with M. Bruguët.

REINHARC Simon, RAFATJAH Matthieu and BRUNET Charles (M1-MIDO) *Study of available bootstrap methods for estimating $MAR(p,q)$ mixed-causal models by (quasi-) maximum likelihood.*

BEAUVARLET Quentin and BLASCO Anaïs (M1-MIDO). *Study of available methods for forecasting $MARMA(p,q,r,s)$ models with α -stable distribution.*

2022. Anastasia Schenkery, Axel Sauvaget and Mathieu Navarro (M1-MIDO), *Study of the predictive density of causal and non-causal ARMA models using machine learning approaches.*

Karen Arban and Axelle Roques (M1-IEF), *Speculative bubbles*

Iannis Reuter (M2-AID), *Measuring the temperature sensitivity of energy consumption* jointly with R. Le Saout (CREST).

MEMBER OF PHD COMMITTEES

2025 Peter MacKenzie (York University, Toronto), *Bridging the digital divide and mitigating cyber security risks in Canada.*

2023 Francesco Giancaterini (Maastricht University), *Essays on Univariate and Multivariate Noncausal Processes.*

COMPUTER SKILLS

Julia (expert), EViews (expert), Gretl, LaTeX, MATLAB (expert), Python (expert), R (expert), SAS (expert), Shiny (advanced), SPARK, STATA, Parallel Computing.

TEACHING

- **Econometrics for Energy Economists (Master 2 EFC, Paris Dauphine 2026-)**
- **Noncausal econometrics (PhD lecture SDOSE, Paris Dauphine and Mines PSL in English, 2024-)**
- **Computer Science Project (Master 2 Digital Economics, in English, 2023-)**
- **Topics in Advanced Economic Analysis (Master 1 QEA in English, 2023-)**
- **Machine Learning for Quantitative Finance (Master 2 272 IEF, 2022-)**
- **Applied Time Series (Master 1 IEF and Master 1 AID, 2022-)**
- **PSL-WEEK : Machine Learning and AI for Economics and Finance (2022)**